

Question 2 Run code

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question
2\sliding_window_max.go'
```

```
[3 3 5 5 6 7]
```

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question 2\group_anagrams.go'

[[eat tea ate] [tan nat] [bat]]
```

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question
2\sliding_window_max.go'
```

```
[3 3 5 5 6 7]
```

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question 2\student_grades.go'
```

```
Top Student: Sara, Average: 96.50
```

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question
2\sort_interface_demo.go'
```

```
Sorted by Age (Ascending):
```

```
Omar - 20
```

```
Ali - 25
```

```
Sara - 30
```

```
Ranked by Score (Descending):
```

```
Sara - 30
```

```
Ali - 25
```

```
Omar - 20
```

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question 2\retry.go'
```

Success!

Failed: after 2 attempts: server unreachable

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question  
2\worker_pool_recover.go'
```

--- Results ---

Job 1 -> Result: 10

Job 2 -> Result: 20

Job 4 -> Result: 40

Job 5 -> Result: 50

Job 7 -> Result: 70

Job 8 -> Result: 80

Job 10 -> Result: 100

--- Errors ---

worker 3 recovered from panic on job 3: simulated crash on job 3

worker 1 recovered from panic on job 6: simulated crash on job 6

worker 2 recovered from panic on job 9: simulated crash on job 9

Program finished successfully without crashing!

```
PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question 2\ticker_rate_limiter.go'
```

Task 1 started at 01:11:11.417

Task 1 finished

Task 2 started at 01:11:11.616

Task 2 finished

Task 3 started at 01:11:11.816

Task 3 finished

Task 4 started at 01:11:12.017

Task 4 finished

Task 5 started at 01:11:12.216

Task 5 finished

Task 6 started at 01:11:12.416

Task 6 finished

Task 7 started at 01:11:12.617

Task 7 finished

Task 8 started at 01:11:12.817

Task 8 finished

Task 9 started at 01:11:13.017

Task 9 finished

Task 10 started at 01:11:13.216

Task 10 finished

Task 11 started at 01:11:13.416

Task 11 finished

Task 12 started at 01:11:13.616

Task 12 finished

Task 13 started at 01:11:13.816

Task 13 finished

Task 14 started at 01:11:14.016

Task 14 finished

Task 15 started at 01:11:14.216

Task 15 finished

Task 16 started at 01:11:14.416

Task 16 finished

Task 17 started at 01:11:14.616

Task 17 finished

Task 18 started at 01:11:14.816

Task 18 finished

Task 19 started at 01:11:15.016

Task 19 finished

Task 20 started at 01:11:15.216

Task 20 finished

All 20 tasks processed through the rate limiter.

PS C:\Users\abouelfetouh\Desktop\Week11_sec1\GoLangExam1> go run '.\Question 2\parallel_sum.go'

Total Sum: 120